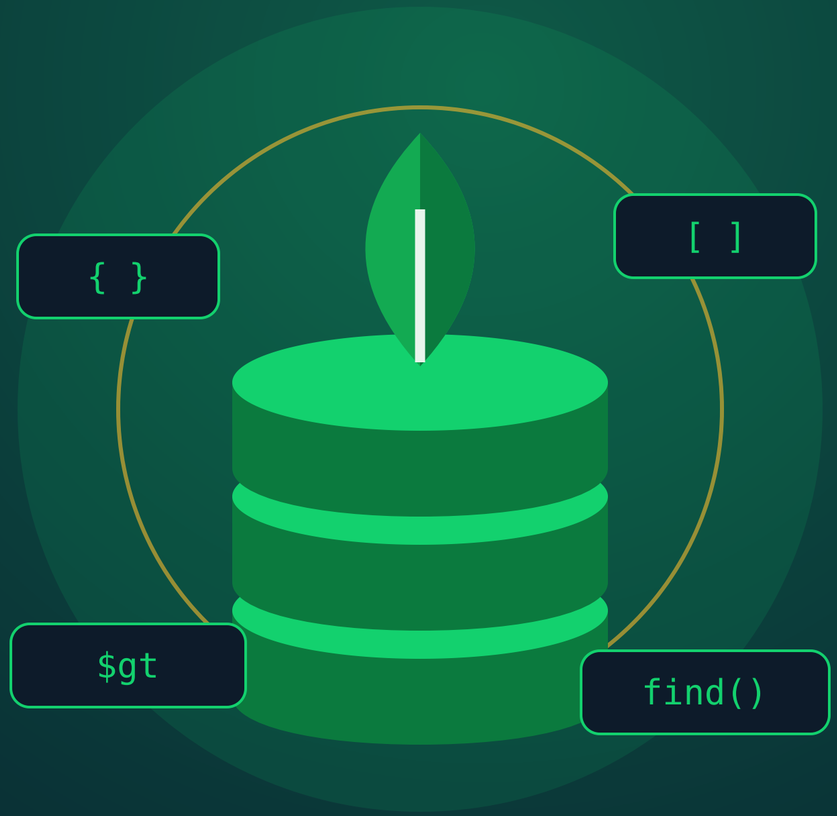
BIG DATA COURSE

SECTION 5

Case Study: Library System

Apply what you learned: insert, search, update and delete.

Prepared by: Amira Mahmoud



SECTION 5

The scenario

The library writes all its books on paper.
The librarian wants to keep the books in a **MongoDB** database.
Your job: answer the librarian's questions using MongoDB.

Add
Put the books in the database

Find
Search for books

Change
Update the number of copies

Delete
Remove a book

SECTION 5

Before you start

1. Open Compass and open the **mongosh** tab
2. Write `use library`
3. Run the insert command on the next slides
4. Check: `db.books.countDocuments()` must give **6**

FIRST COMMAND

```
use library
```

SECTION 5

The data: 6 books

Title	Author	Year	Rating	Copies	Type	Publisher
Palace Walk	Naguib Mahfouz	1956	4.8	3	novel, history	Cairo Books (Egypt)
The Days	Taha Hussein	1929	4.7	2	biography, history	Nile Press (Egypt)
Return of the Spirit	Tawfiq al-Hakim	1933	4.5	4	novel, drama	Beirut Books (Lebanon)
Utopia	Ahmed Khaled Towfik	2008	4.0	1	novel, science fiction	Cairo Books (Egypt)
The Yacoubian Building	Alaa Al Aswany	2002	3.9	2	novel, drama	Beirut Books (Lebanon)
Children of the Alley	Naguib Mahfouz	1959	4.6	2	novel, history	Nile Press (Egypt)

Arabic books. `genres` is a list. `publisher` is a small document inside the book.

SECTION 5

Insert the books

RUN IN THE SHELL · after use library

```
db.books.drop()
db.books.insertMany([
  { title: "Palace Walk", author: "Naguib Mahfouz", year: 1956, rating: 4.8, copies: 3, genres: ["novel", "history"], publisher: {
name: "Cairo Books", country: "Egypt" } },
  { title: "The Days", author: "Taha Hussein", year: 1929, rating: 4.7, copies: 2, genres: ["biography", "history"], publisher: { name:
"Nile Press", country: "Egypt" } },
  { title: "Return of the Spirit", author: "Tawfiq al-Hakim", year: 1933, rating: 4.5, copies: 4, genres: ["novel", "drama"],
publisher: { name: "Beirut Books", country: "Lebanon" } },
  { title: "Utopia", author: "Ahmed Khaled Towfik", year: 2008, rating: 4.0, copies: 1, genres: ["novel", "science fiction"],
publisher: { name: "Cairo Books", country: "Egypt" } },
  { title: "The Yacoubian Building", author: "Alaa Al Aswany", year: 2002, rating: 3.9, copies: 2, genres: ["novel", "drama"],
publisher: { name: "Beirut Books", country: "Lebanon" } },
  { title: "Children of the Alley", author: "Naguib Mahfouz", year: 1959, rating: 4.6, copies: 2, genres: ["novel", "history"],
publisher: { name: "Nile Press", country: "Egypt" } }
])
```

SECTION 5

The librarian's questions (1 to 4)

1. **How many books are in the library?**
   Hint: count

2. **Show all drama books.**
   Hint: the list contains drama

3. **Show old books (before year 2000) with rating 4.7 or more.**
   Hint: two conditions: less than, and greater or equal

4. **Show the books from a publisher in Lebanon.**
   Hint: publisher.country

SECTION 5

The librarian's questions (5 to 7)

5 **Show the 3 best rated books (title and rating only).**

Hint: sort and limit

6 **A new copy of "Palace Walk" came. Add 1 to its copies.**

Hint: updateOne with $inc

7 **"Utopia" is removed from the library. Delete it. How many books are left?**

Hint: deleteOne

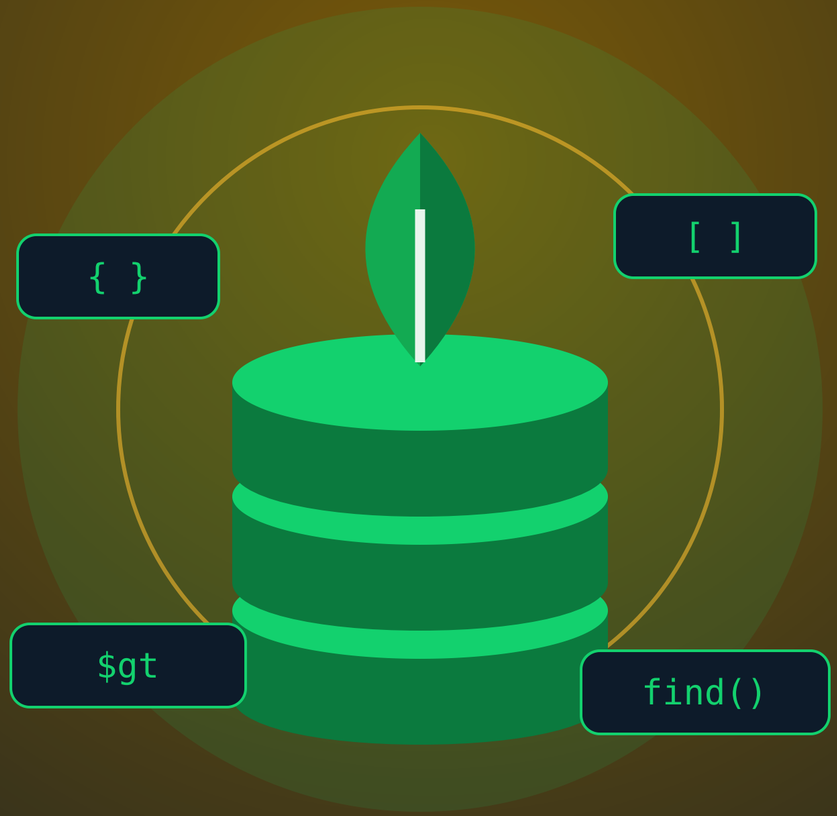
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SECTION 5

Solutions

Answers and results for the 7 questions

Prepared by: Amira Mahmoud



SOLUTION

Solution: insert and count (task 1)

COMMAND

```
use library
db.books.countDocuments()
```

mongosh

```
test> use library
switched to db library
test> db.books.countDocuments()
6
```

Example output

SOLUTION

Solution: drama books (task 2)

COMMAND

```
db.books.find(
  { genres: "drama" },
  { title: 1, _id: 0 }
)
```

mongosh

```
library> db.books.find({ genres: "drama" }, { title: 1, _id: 0 })
[ { title: 'Return of the Spirit' }, { title: 'The Yacoubian
Building' } ]
```

Example output

`genres` is a list. This filter means: the list contains drama.

SOLUTION

Solution: old and highly rated (task 3)

COMMAND

```
db.books.find(
  { year: { $lt: 2000 }, rating: { $gte: 4.7 } },
  { title: 1, year: 1, rating: 1, _id: 0 }
)
```

mongosh

```
library> db.books.find({ year: { $lt: 2000 }, rating: { $gte: 4.7 }
}, ...)
[
  { title: 'Palace Walk', year: 1956, rating: 4.8 },
  { title: 'The Days', year: 1929, rating: 4.7 }
]
```

Example output

Two conditions in one filter: both must be true.

SOLUTION Solution: publisher in Lebanon (task 4)

COMMAND

```
db.books.find(
  { "publisher.country": "Lebanon" },
  { title: 1, "publisher.country": 1, _id: 0 }
)
```

mongosh

```
library> db.books.find({ "publisher.country": "Lebanon" }, ...)
[
  { title: 'Return of the Spirit', publisher: { country: 'Lebanon' }
},
  { title: 'The Yacoubian Building', publisher: { country: 'Lebanon'
} }
]
```

Example output

> Use a dot to enter the document inside. Keep it in quotes.

SOLUTION

Solution: top 3 rated books (task 5)

COMMAND

```
db.books.find(
  {},
  { title: 1, rating: 1, _id: 0 }
).sort({ rating: -1 }).limit(3)
```

mongosh

```
library> db.books.find({}, { title: 1, rating: 1, _id: 0 }).sort({
rating: -1 }).limit(3)
[
  { title: 'Palace Walk', rating: 4.8 },
  { title: 'The Days', rating: 4.7 },
  { title: 'Children of the Alley', rating: 4.6 }
]
```

Example output

`{}` means no condition. `-1` means highest first. `limit(3)` means only 3.

SOLUTION

Solution: a new copy of Palace Walk (task 6)

COMMANDS

```
db.books.updateOne(
  { title: "Palace Walk" },
  { $inc: { copies: 1 } }
)

db.books.findOne(
  { title: "Palace Walk" },
  { title: 1, copies: 1, _id: 0 }
)
```

mongosh

```
library> db.books.updateOne({ title: "Palace Walk" }, { $inc: {
copies: 1 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library> db.books.findOne({ title: "Palace Walk" }, { title: 1,
copies: 1, _id: 0 })
{ title: 'Palace Walk', copies: 4 }
```

Example output

SOLUTION

Solution: delete Utopia (task 7)

COMMANDS

```
db.books.deleteOne({ title: "Utopia" })

db.books.countDocuments()
```

You cannot undo a delete. 6 books minus 1 = 5.

mongosh

```
library> db.books.deleteOne({ title: "Utopia" })
{ acknowledged: true, deletedCount: 1 }
library> db.books.countDocuments()
5
```

Example output

SOLUTION

What each task practiced

Question	Idea	Command
1	Count	`countDocuments()`
2	Search in a list	`{ genres: "drama" }`
3	Comparison and AND	`$lt` , `$gte`
4	Field inside a document	`"publisher.country"`
5	Sort and limit	`sort()` , `limit()`
6	Update a number	`updateOne` + `$inc`
7	Delete	`deleteOne`

BIG DATA COURSE

SECTION 5

Thank you

Well done!

Prepared by: Amira Mahmoud

